John P. Acosta

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY Newark, NJ

Bachelor of Science in Computer Science, Ying Wu College of Computing *Expected May 2027*

- **GPA:** 3.7 **Awards:** NSF S-STEM Scholar, Dean's List (All Semesters), Ron Andrews PGIM Scholar, Kupfrian Memorial Scholar
- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Design & Management, Operating Systems, Computer Networks, Computer Systems, Artificial Intelligence, Programming Language Concepts (C/C++)

SKILLS

Languages: Python, Java, C, C++, TypeScript, Bash

Systems & Infrastructure: Linux, Docker, Terraform, SLURM/HPC, AWS, CloudFormation

Build, CI/CD & Development: GitHub Actions, Git/GitHub, pytest, FastAPI, React/Next.js, PostgreSQL

AI/ML & Frameworks: PyTorch, Transformers, PEFT, LangChain, GraphRAG, NumPy, pandas

PROFESSIONAL EXPERIENCE

Prudential PGIM Technology Jun 2026 - Aug 2026

Software Engineer Intern

- Developed **7 reusable Terraform modules** for AWS Lambda, S3, and KMS to migrate off CloudFormation, cutting average build time for pilot applications by ~**66%** and infrastructure code by ~**80%**
- Automated reporting on **$1M+** in monthly AWS spend via a React and FastAPI dashboard, replacing a **3-hour manual process**
- Led a 5-engineer team to **2nd place** in PGIM's internal hackathon with an AI-powered Confluence pipeline that detects stale pages, suggests updates, and notifies reviewers, improving the documentation Copilot and Claude MCP rely on

NJIT – Lijing Wang Lab Jan 2026 - Present

Undergraduate Research Assistant

- Turned a professor's research hypothesis on gradient cancellation into a modular Python framework for rapid iteration, executing **250+ controlled fine-tuning runs** across 4 language and vision models on an A100 GPU cluster (**6,000+ GPU service units**)
- Audited Hugging Face and PyTorch internals to remove hidden global seed state, building custom seed modules and checkpoints so each source of randomness could be tested in isolation, preventing invalidation of **400+ planned runs**

NJIT – AI Imaging Medicine Lab Jan 2026 - Jun 2026

Applied AI Research Engineer Intern

- Ported a neurosurgical PyQt application to Linux HPC and built an automated QA harness over 100 natural-language prompt suites, capturing UI screenshots and telemetry to isolate failures, driving fixes that raised pass rates from **20% to 100%**
- Designed a LangChain workflow that breaks one-shot tumor diagnosis into a **5-step** radiologist-style reasoning chain (e.g., symmetry, lesion location), each a dedicated model call on BraTS MRI data, delivering a prototype to inform the lab's system design

NJIT – Lijing Wang Lab Aug 2025 - Jan 2026

Deep Learning Research Intern

- Parallelized independent forecasting workloads across CPU cores with joblib, cutting experiment runtime by ~**80%** and enabling faster iteration without HPC resources
- Engineered modular Python preprocessing (NumPy, pandas) to clean, trim, and interpolate **10+ multi-location epidemic datasets** (~900 weekly observations each), producing consistent train/test splits for rolling-origin forecasting across 5 diseases

TECHNICAL PROJECTS

REVIEW LENS AI | FastAPI, Next.js, PostgreSQL, pgvector, LangChain, GitHub Actions

- Designed a restaurant-review Q&A agent with deterministic safety guardrails, combining a SQL-built dataset summary, predefined SQL query tools, and pgvector semantic search, with data access restricted at the tool level, preventing cross-user access
- Built a fault-tolerant data ingestion pipeline loading restaurant reviews into Postgres (FastAPI, background worker) with a run state machine, a database-enforced one-active-run constraint, transactional stage checkpoints for safe retries, and PII scrubbing
- Built GitHub Actions CI (lint, type checks, unit and Postgres integration tests) with scripted branch protection, and migrated the frontend to OpenAPI-generated clients and TanStack Query hooks via Orval

AI-TUTOR | Python, GraphRAG, Docker, Flask, BeautifulSoup

- Built a **knowledge-graph tutoring system** that grounds LLM responses in retrieved context, using a custom BFS web crawler and an extended nano-graphrag implementation with domain-specific entity extraction
- Implemented rate-limited LLM requests with exponential backoff and integrated a Flask backend with a Streamlit interface in a Dockerized development environment supporting selective service loading

EXTRACURRICULAR ACTIVITIES

Society of Hispanic Professional Engineers (SHPE) | NJIT Chapter | Member *Jan 2024 - Present*

HISPA (Hispanics Inspiring Students' Performance and Achievements) | Member & Role Model *Sep 2025 - Present*

Break Through Tech AI Fellow | Cornell Tech | Machine Learning Certificate *Jun 2025 - Present*

Google Software Engineering Fellow (Selected 2x) | G-SWEP *Jul 2025 - Sep 2025*